

Element K

We were tasked with creating a large rose for the “Beauty and the Beast” musical. Mrs. Moody required that the rose drop its petals at three specific times throughout the musical. She wanted the rose to be at least 1.5 ft tall and look as close to a rose as possible to express a magical feeling to the audience. The rose also needs to have lights shining from under the rose. Each member of our group came up with at least five different prototypes to get the most ideas. We created a design matrix that ranked the different designs based on the criteria to be created. The criteria was weighted to give more importance to the criteria we thought was most important to the completion of our task. We ended up choosing a design that used strings to hold the petals in place and drop the petals by unwinding the string on a spool in the base of the rose. We made a list of materials we needed and ordered the ones we did not already have. Our prototype used a string that was tied to a real rose petal. The string was threaded through a tube. The prototype showed how the petal would be lowered and could be reset easily. We got advice from students and professors from Vanderbilt University. They advised us not to use a controller to make it simpler. They also gave us advice on how to build it to make it look as close to a rose as possible. We started by making the stem out of pvc. We put three thin tubes in the PVC that each had a fishing line through them. The fishing line was attached to three different spools that each had its own motor. The motors were obtained from rc cars that had controllers. The petals were made out of felt and red cloth with a wire frame. The stem was painted green, and the base was painted black. We tested the system of motors and found they were too fast and the fishing line did not unwind well. The petals were too heavy. We found when we wound the fishing line up, the petals would flop down and not stand up straight. We rebuilt the rose and used a thinner stem made of a tube and masking tape. We also changed the bulb of the flower to one of the fake flowers we got at the beginning. This made our rose realistic. Our product had some good parts but also had many problems. First, the main system of lowering the petals did not work well. The mechanism only lowered one petal. One good part about the product is the controller worked well and could control all three motors. The lights also did a good job of providing light for the rose. One of the main things we learned is the importance of testing parts as we go. It would have been better to make changes to the rose to fix problems instead of building the whole thing before testing. This made it hard to fix problems without taking apart the whole rose. We also learned it is better to get better criteria from the stakeholders. The information from Mrs. Moody was vague and she left us with little guidance on how to make it. When we showed it to her, she was unhappy with the way it looked. It would have been better to ask better questions to find what she really wanted.